

Experiment 2

Aim: To demonstrate 2 class confusion matrix using python

Program

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import
accuracy_score,f1_score,precision_score,recall_score,classification_report,confusion_matrix
wine=load_wine()
data=pd.DataFrame(data=wine.data,columns=wine.feature_names)
data['Target']=wine.target
data=data[data['Target']!=2]
x=data.drop('Target',axis=1)

y=data['Target']
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3,random_state=1)
model=DecisionTreeClassifier(random_state=1)
model.fit(x_train,y_train)
y_pred=model.predict(x_test)
accuracy=accuracy_score(y_test,y_pred)
print("accuracy:",accuracy)
class_report = classification_report(y_test, y_pred,
target_names=wine.target_names[:2])
print("Classification Report:\n", class_report)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
print(f'Precision: {precision:.2f}')
print(f'Recall: {recall:.2f}')
```

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print(f"F1 Score: {f1:.2f}")
conf_matrix=confusion_matrix(y_test,y_pred)
plt.figure(figsize=(8,6))
sns.heatmap(conf_matrix,annot=True,fmt='d',cmap='PuBuGn',xticklabels=wine.
target_names[:2],yticklabels=wine.target_names[:2])
plt.xlabel("predicted label")
plt.ylabel("true label")
plt.title("confusion matrix")
plt.show()

```

Output:

```

***  accuracy: 0.9743589743589743
      Classification Report:
              precision    recall  f1-score   support

   class_0       1.00      0.94      0.97        17
   class_1       0.96      1.00      0.98        22

   accuracy          0.98
  macro avg          0.98
 weighted avg          0.98

Precision: 0.96
Recall: 1.00
F1 Score: 0.98

```

